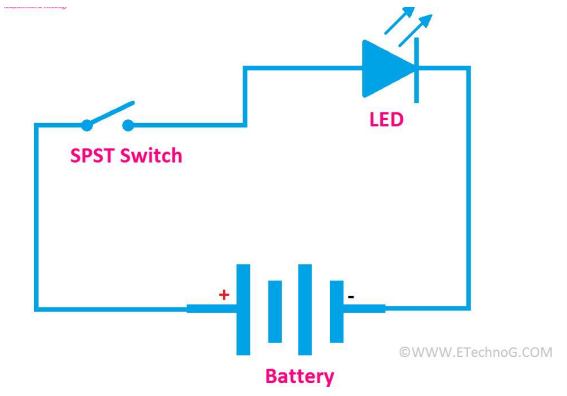
Switches and Relay

Mechanical Switches

- A mechanical device that opens or closes a circuit.
- Used to **control the flow** of current.

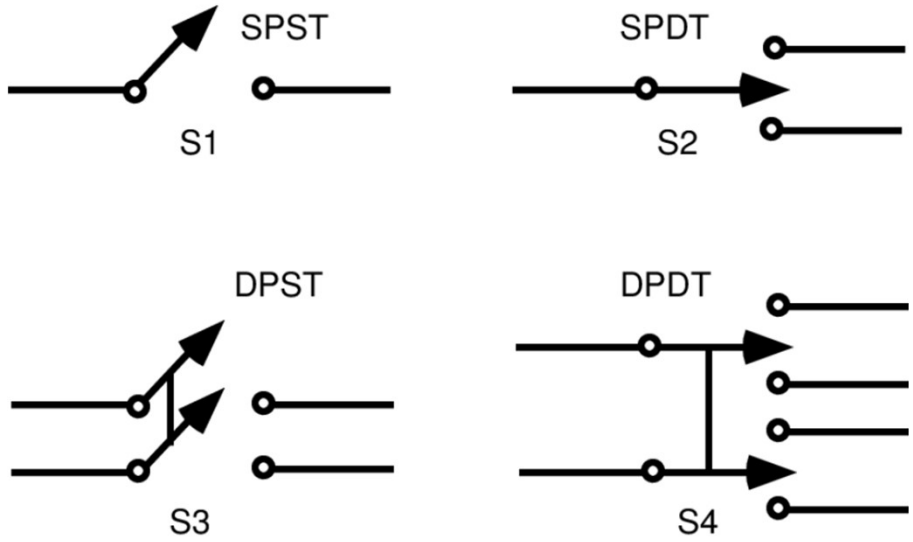


Types of Switches Examples:
SPST → one line open/close
SPDT → one line switching between two

Pole: The number of separate circuits a switch can control (Input connection)
Throw: A switch throw describes the number of output connections each pole of the switch can hav

Types of Switches

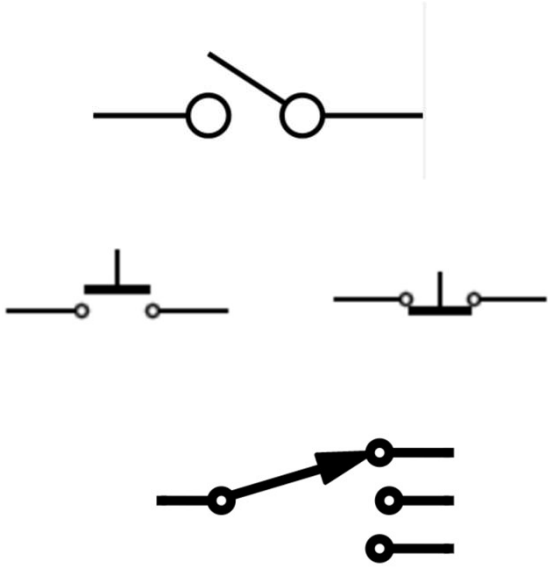
SPST → one line open/close
SPDT → one line switching between two
DPST → two lines ON/OFF together
DPDT → two lines switching between two



Pole: The number of separate circuits a switch can control (Input connection)
Throw: A switch throw describes the number of output connections each pole of the switch can have

Types Mechanical Switches

- **Toggle Switch:** A switch with a lever that moves back and forth to control a circuit.
- **Push Button Switch:** Momentary or maintained contact, commonly used in calculators and phones.
- **Rotary Switch:** Used to select one of several circuit paths, found in volume controls.
- **Slide Switch:** Sliding motion to open/close circuits, found in smaller electronics.

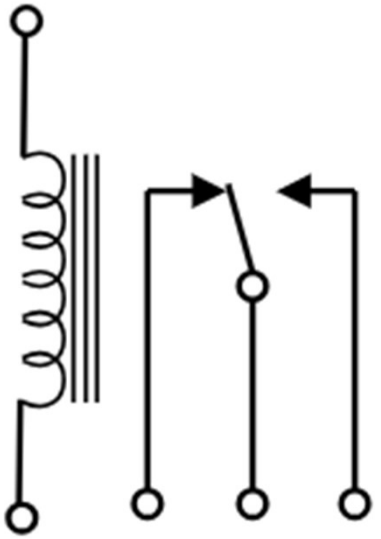


Electromagnetic Relay

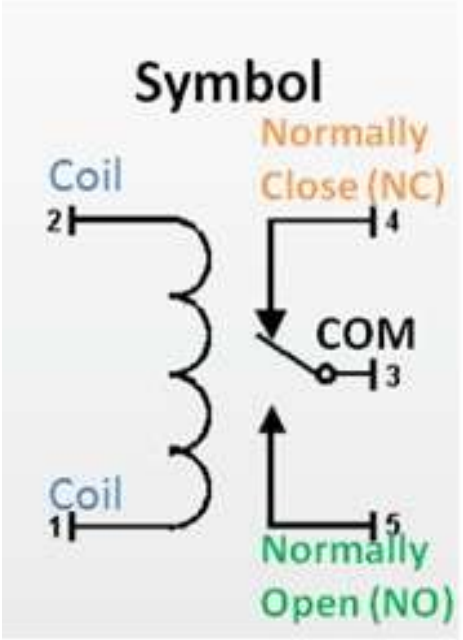
- An electromagnetic switch that uses a **small control current** to switch a **larger current**.
- Provides electrical isolation between control and power circuits.

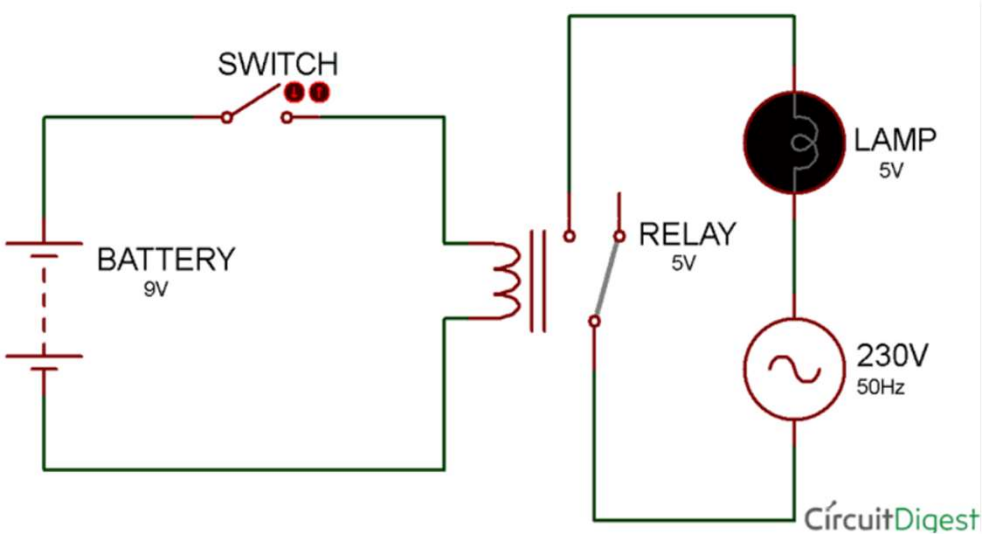
Coil → creates a magnetic field when energized

Contacts → open/close the circuit



A simple Relay Circuit





PC: https://circuitdigest.com/